

# Modelling and simulation

**Ayush Yadav 1836420**

**Vinay Aditya Thirunagari 1830650**

Output

(30.333905254374553, 1, 'Arriving', 0)  
(95.96165995999333, 2, 'Arriving', 1)  
(147.0213580878296, 1, 'Testing', 2)  
(208.97160218664393, 3, 'Arriving', 2)  
(295.57534950775414, 4, 'Arriving', 3)  
(356.29804844866953, 5, 'Arriving', 4)  
(410.66394742428395, 6, 'Arriving', 5)  
(482.5754266727015, 7, 'Arriving', 6)  
(580.4693286120746, 8, 'Arriving', 7)  
(638.5730089097083, 9, 'Arriving', 8)  
(669.7351524508912, 10, 'Arriving', 9)  
(704.7865939708186, 11, 'Arriving', 10)  
(820.0375470617457, 12, 'Arriving', 11)  
(891.3400084608102, 13, 'Arriving', 11)  
(960.3387356017851, 14, 'Arriving', 11)  
(1032.6140292231335, 15, 'Arriving', 11)  
(1118.4600255247333, 16, 'Arriving', 11)  
(1217.5446353855443, 17, 'Arriving', 11)

(1283.7120781652125, 18, 'Arriving', 11)  
(1347.0213580878296, 1, 'Leaving', 11)  
(1357.0003663532964, 19, 'Arriving', 10)  
(1465.931426156998, 2, 'Testing', 11)  
(1473.4878404703388, 20, 'Arriving', 11)  
(1554.7826893836607, 21, 'Arriving', 11)  
(1644.0979535260544, 22, 'Arriving', 11)  
(1739.9925992576875, 23, 'Arriving', 11)  
(1846.943677040547, 24, 'Arriving', 11)  
(1936.721828167772, 25, 'Arriving', 11)  
(2019.6199383299002, 26, 'Arriving', 11)  
(2074.911366106038, 27, 'Arriving', 11)  
(2140.154751721585, 28, 'Arriving', 11)  
(2223.2076935990995, 29, 'Arriving', 11)  
(2273.7521161840127, 30, 'Arriving', 11)  
(2383.8336734435575, 31, 'Arriving', 11)  
(2458.81549795826, 32, 'Arriving', 11)  
(2542.9876301308545, 33, 'Arriving', 11)  
(2660.4784492786234, 34, 'Arriving', 11)  
(2665.931426156998, 2, 'Leaving', 11)  
(2761.9913562644047, 35, 'Arriving', 10)  
(2784.7701187317366, 3, 'Testing', 11)  
(2832.639964063859, 36, 'Arriving', 11)  
(2941.9016962706514, 37, 'Arriving', 11)  
(3022.930709322521, 38, 'Arriving', 11)  
(3134.390189299642, 39, 'Arriving', 11)  
(3234.4187176315377, 40, 'Arriving', 11)  
(3327.2368011893163, 41, 'Arriving', 11)

(3398.34952720008, 42, 'Arriving', 11)  
(3449.662427230792, 43, 'Arriving', 11)  
(3496.025114641394, 44, 'Arriving', 11)  
(3579.531966604504, 45, 'Arriving', 11)  
(3623.0104876221662, 46, 'Arriving', 11)  
(3673.2557543395615, 47, 'Arriving', 11)  
(3732.4463503900984, 48, 'Arriving', 11)  
(3744.7701187317366, 3, 'Leaving', 11)  
(3798.2509589597576, 49, 'Arriving', 10)  
(3857.297450164766, 4, 'Testing', 11)  
(3905.4376733504064, 50, 'Arriving', 11)  
(4017.379502262932, 51, 'Arriving', 11)  
(4066.752053355901, 52, 'Arriving', 11)  
(4183.525812134919, 53, 'Arriving', 11)  
(4280.233637703163, 54, 'Arriving', 11)  
(4366.624161022847, 55, 'Arriving', 11)  
(4461.762221517829, 56, 'Arriving', 11)  
(4575.002627795426, 57, 'Arriving', 11)  
(4624.360115007074, 58, 'Arriving', 11)  
(4701.826528443914, 59, 'Arriving', 11)  
(4788.001847874206, 60, 'Arriving', 11)  
(4817.297450164766, 4, 'Leaving', 11)  
(4902.961848967903, 61, 'Arriving', 10)  
(4910.008255767575, 5, 'Testing', 11)  
(4978.984454160174, 62, 'Arriving', 11)  
(5016.7196417102095, 63, 'Arriving', 11)  
(5060.8531824141155, 64, 'Arriving', 11)  
(5157.46992721955, 65, 'Arriving', 11)

(5215.922286636503, 66, 'Arriving', 11)  
(5322.819847708624, 67, 'Arriving', 11)  
(5356.59701577773, 68, 'Arriving', 11)  
(5402.794708547142, 69, 'Arriving', 11)  
(5493.644112433202, 70, 'Arriving', 11)  
(5581.54655206739, 71, 'Arriving', 11)  
(5612.032835482326, 72, 'Arriving', 11)  
(5630.008255767575, 5, 'Leaving', 11)  
(5683.271563557591, 73, 'Arriving', 10)  
(5691.154846426893, 6, 'Testing', 11)  
(5770.905833346572, 74, 'Arriving', 11)  
(5802.266831029338, 75, 'Arriving', 11)  
(5916.262043584377, 76, 'Arriving', 11)  
(5976.24474766713, 77, 'Arriving', 11)  
(6012.194345267385, 78, 'Arriving', 11)  
(6104.674498129385, 79, 'Arriving', 11)  
(6174.101765281037, 80, 'Arriving', 11)  
(6208.16694651933, 81, 'Arriving', 11)  
(6263.653275504296, 82, 'Arriving', 11)  
(6352.2132267494235, 83, 'Arriving', 11)  
(6390.516445725539, 84, 'Arriving', 11)  
(6497.654055939207, 85, 'Arriving', 11)  
(6604.37558382047, 86, 'Arriving', 11)  
(6651.154846426893, 6, 'Leaving', 11)  
(6720.021412116695, 87, 'Arriving', 10)  
(6747.803098350872, 7, 'Testing', 11)  
(6774.494731581638, 88, 'Arriving', 11)  
(6882.459010573167, 89, 'Arriving', 11)

(6985.110124743488, 90, 'Arriving', 11)  
(7072.735164637854, 91, 'Arriving', 11)  
(7174.361326271707, 92, 'Arriving', 11)  
(7227.803098350872, 7, 'Leaving', 11)  
(7336.976544685837, 8, 'Testing', 10)  
(8536.976544685836, 8, 'Leaving', 10)  
(8617.232927143663, 9, 'Testing', 9)  
(9577.232927143663, 9, 'Leaving', 9)  
(9639.756951918398, 10, 'Testing', 8)  
(10839.756951918398, 10, 'Leaving', 8)  
(10941.99011220684, 11, 'Testing', 7)  
(11181.99011220684, 11, 'Leaving', 7)  
(11291.929936904631, 19, 'Testing', 6)  
(12491.929936904631, 19, 'Leaving', 6)  
(12580.670457055541, 35, 'Testing', 5)  
(13540.670457055541, 35, 'Leaving', 5)  
(13645.873352868897, 49, 'Testing', 4)  
(14845.873352868897, 49, 'Leaving', 4)  
(14944.177627275538, 61, 'Testing', 3)  
(15424.177627275538, 61, 'Leaving', 3)  
(15485.147783890947, 73, 'Testing', 2)  
(16685.14778389095, 73, 'Leaving', 2)  
(16794.52781230767, 87, 'Testing', 1)  
(17274.52781230767, 87, 'Leaving', 1)

Total cars missed: 75

Average people per car: 3.17

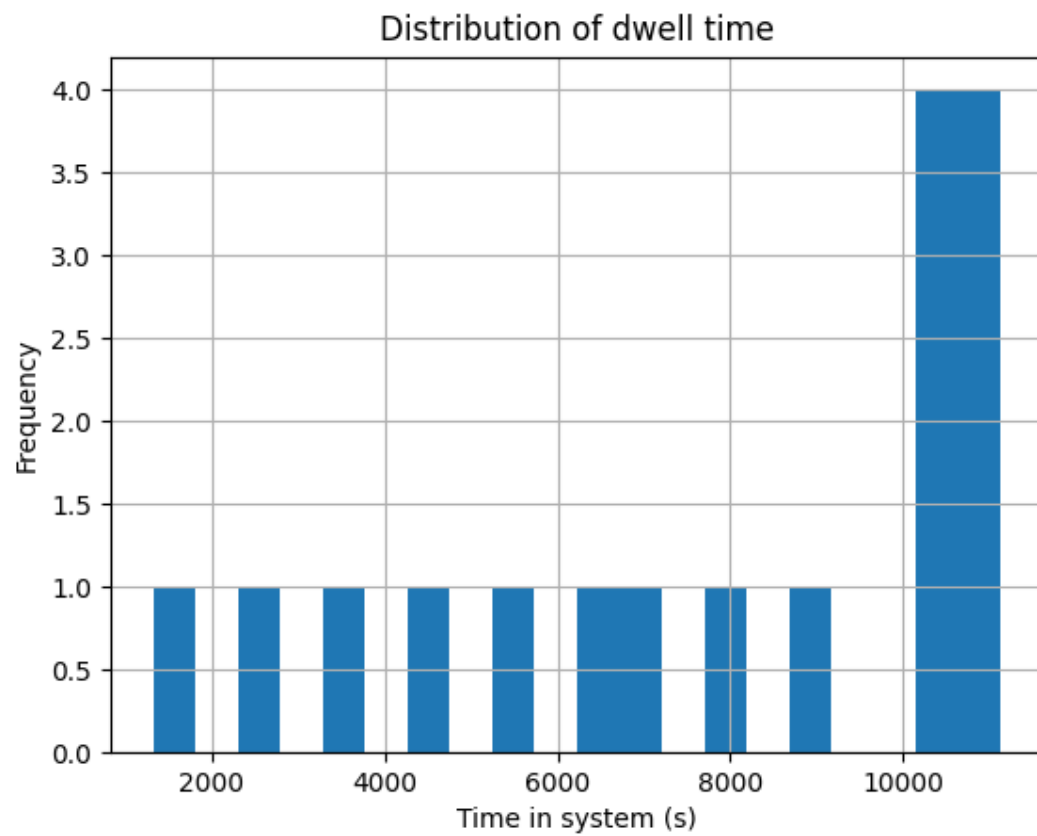
Average cars in system: 7.69

Capacity 10: cars missed = 72

Capacity 12: cars missed = 73

Capacity 14: cars missed = 70

Capacity 16: cars missed = 63



Vehicles in system over time

